

Revisiting SSL/TLS Implementations: New Bleichenbacher Side Channels and Attacks

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As a countermeasure against the famous Bleichenbacher attack on RSA based ciphersuites, all TLS RFCs starting from RFC 2246 (TLS 1.0) propose “to treat incorrectly formatted messages in a manner indistinguishable from correctly formatted RSA blocks”.

In this paper we show that this objective has not been achieved yet (cf. Table 1): We present four new Bleichenbacher side channels, and three successful Bleichenbacher attacks against the *Java Secure Socket Extension (JSSE)* SSL/TLS implementation and against hardware security appliances using the *Cavium NITROX SSL accelerator chip*. Three of these side channels are timingbased, and two of them provide the first timing-based Bleichenbacher attacks on SSL/TLS described in the literature. Our measurements confirmed that all these side channels are observable over a switched network, with timing differences between 1 and 23 microseconds. We were able to

successfully recover the PreMasterSecret using three of the four side channels in a realistic measurement setup.

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